

Jiawei ZHANG

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EDUCATION

University of Chicago

Ph.D. in Computer Science, 4.0/4.0

Advisor: *Prof. Bo Li*

Sept. 2024 – Aug. 2026

University of Illinois Urbana-Champaign

Ph.D. in Computer Science, 4.0/4.0

Advisor: *Prof. Bo Li*

Aug. 2023 – May. 2024

M.S. in Computer Science

May. 2023

Zhejiang University

Bachelor of Engineering

Hangzhou, China

Sept. 2017 – Jun. 2021

RESEARCH INTEREST:

I study **Safe AGI**, with a focus on **making LLMs and LLM agents reliable at scale**. My research runs as a closed loop: (i) **scalable red-teaming** to elicit realistic, long-horizon failures. I participated in internal red-teaming evaluations of **OpenAI o1**, **Google DeepMind**, and **ElevenLabs TTS** with **Virtue AI**. In parallel, I study **agentic reliability** for LLM agents, focusing on agent-specific vulnerabilities (e.g., prompt injection and malicious action execution) as well as **environment security construction** for agentic systems; (ii) **defense & alignment** that transforms safety requirements into actionable safeguards, spanning **certified robustness** for worst-case guarantees and **interpretability-guided defenses** that translate mechanistic insights into generalizable mitigations; and (iii) **better safety-aware reasoning for autonomous driving** by incorporating richer world knowledge via reinforcement learning for vision–language–action models.

Together, I aim to build **scalable red-teaming pipelines for LLM agents** and develop **practical patches** that improve safety awareness **while maintaining scalability**. Previously, I was an **Anthropic Fellow for AI Safety Research** from January to July 2026.

SELECTED PUBLICATION (* DENOTES CO-FIRST AUTHORSHIP)

- **Jiawei Zhang**, Andrew Estornell, David D. Baek, Bo Li, Xiaojun Xu. Any-Depth Alignment: Unlocking Innate Safety Alignment of LLMs to Any-Depth. *International Conference on Learning Representations (ICLR) 2026*. [arXiv]
- Zhaorun Chen*, Xun Liu*, Mintong Kang, **Jiawei Zhang**, Minzhou Pan, Shuang Yang, Bo Li. ARMs: Adaptive Red-Teaming Agent against Multimodal Models with Plug-and-Play Attacks. *International Conference on Learning Representations (ICLR) 2026*. [arXiv]
- Mintong Kang*, Zhaorun Chen*, Chejian Xu*, **Jiawei Zhang***, Chengquan Guo*, Minzhou Pan, Ivan Revilla, Yu Sun, Bo Li. GuardSet-X: Massive Multi-Domain Safety Policy-Grounded Guardrail Dataset. *NeurIPS 2025*. [arXiv]
- **Jiawei Zhang**, Shuang Yang, Bo Li. UDora: A Unified Red Teaming Framework Against LLM Agents by Dynamically Leveraging Their Own Reasoning. *International Conference on Machine Learning (ICML) 2025*. [arXiv]
- **Jiawei Zhang**, Xuan Yang, Taiqi Wang, Yu Yao, Aleksandr Petiushko, Bo Li. SafeAuto: Knowledge-Enhanced Safe Autonomous Driving with Multimodal Foundation Models. *International Conference on Machine Learning (ICML) 2025*. [arXiv]
- Chejian Xu*, **Jiawei Zhang***, Zhaorun Chen*, Chulin Xie*, Mintong Kang*, Zhuowen Yuan*, Chenhui Zhang, Lingzhi Yuan, Yi Zeng, Peiyang Xu, Chengquan Guo, Andy Zhou, ..., Zidi Xiong, Zinan Lin, Dan Hendrycks, Dawn Song, Bo Li. MMDT: Decoding the Trustworthiness and Safety of Multimodal Foundation Models. *International Conference on Learning Representations (ICLR) 2025*. [arXiv]
- **Jiawei Zhang**, Chejian Xu, Yu Gai, Freddy Lecue, Dawn Song, Bo Li. KnowHalu: Hallucination Detection via Multi-Form Knowledge Based Factual Checking. *ICLR 2025 Workshop on Foundation Models in the Wild*. [arXiv]
- Bowen Jin, Chulin Xie, **Jiawei Zhang**, Kashob Kumar Roy, Yu Zhang, Zheng Li, Ruirui Li, Xianfeng Tang, Suhang Wang, Yu Meng, Jiawei Han. Graph Chain-of-Thought: Augmenting Large Language Models by Reasoning on Graphs. *Findings of the Association for Computational Linguistics (ACL) 2024*. [arXiv]

- **Jiawei Zhang**, Chejian Xu, Bo Li. ChatScene: Knowledge-Enabled Safety-Critical Scenario Generation for Autonomous Vehicles. *Conference on Computer Vision and Pattern Recognition (CVPR) 2024*. [arXiv]
- **Jiawei Zhang**, Tianyu Pang, Chao Du, Yi Ren, Bo Li, Min Lin. MMCBench: Benchmarking Large Multimodal Models against Common Corruptions. [arXiv]
- **Jiawei Zhang**, Zhongzhu Chen, Huan Zhang, Chaowei Xiao, Bo Li. DiffSmooth: Certifiably Robust Learning via Diffusion Models and Local Smoothing. *32th USENIX Security Symposium 2023*. [arXiv]
- **Jiawei Zhang**, Linyi Li, Ce Zhang, Bo Li. CARE: Certifiably Robust Learning with Reasoning via Variational Inference. *IEEE Conference on Secure and Trustworthy Machine Learning (SatML) 2023*. [arXiv]
- Zhuolin Yang*, Zhikuan Zhao*, Boxin Wang, **Jiawei Zhang**, Linyi Li, Hengzhi Pei, Bojan Karlas, Ji Liu, Heng Guo, Ce Zhang, Bo Li. Improving Certified Robustness via Statistical Learning with Logical Reasoning. *Advances in Neural Information Processing Systems (NeurIPS) 2022*. [arXiv]
- Linyi Li, **Jiawei Zhang**, Tao Xie, Bo Li. Double Sampling Randomized Smoothing. *International Conference on Machine Learning (ICML) 2022*. [arXiv]
- **Jiawei Zhang***, Linyi Li*, Huichen Li, Xiaolu Zhang, Shuang Yang, Bo Li. Progressive-Scale Boundary Blackbox Attack via Projective Gradient Estimation. *International Conference on Machine Learning (ICML) 2021*. [arXiv]

INDUSTRY RESEARCH EXPERIENCE

Anthropic

AI Safety Research Fellow

Jan 2026 – July 2026

- Automatic generation of environments and attacks for prompt injection evaluation at scale.

NVIDIA

ML Research Intern, Santa Clara

Advised by Dr. Wenhao Ding, Dr. Yulong Cao, Dr. Boris Ivanovic, and Prof. Marco Pavone

Oct 2025 – Jan 2026

- Explored improved reinforcement learning methods for reasoning to reduce minADE.
- Developed enhanced reasoning capabilities for autonomous driving by leveraging world models.

ByteDance Seed Research

Responsible AI Research Intern, San Jose

Advised by Dr. Xiaojun Xu, Dr. Andrew Estornell, and Dr. Hang Li

June 2025 – Sept 2025

- Developed a scalable, model-agnostic defense for LLMs based on safety-signature tokens, effective against deep prefill attacks with context lengths exceeding 3,000 tokens.
- Demonstrated robustness to in-the-wild jailbreaks (AutoDAN, PAIR) while maintaining 0% false positive rate on standard benchmarks (MMLU, MATH, BBH, AlphaEval).

Nuro AI

Machine Learning Research Intern (Pathfinder), Mountain View

Advised by Dr. Yu Yao, Taiqi Wang, and Dr. Aleksandr Petiushko

May 2024 – Aug 2024

- Fine-tuned a multimodal LLM for video-conditioned autonomous driving to output high-level action plans and low-level control signals, unifying perception, reasoning, and control.
- Developed a multimodal retrieval-augmented training pipeline and a Markov Logic Network that encodes first-order traffic rules to verify and improve the safety of LLM-proposed actions.
- Proposed a position-dependent cross-entropy loss (PDCE) to stabilize and improve numeric control predictions when represented as text.

Sea AI Lab

Machine Learning Research Intern, Singapore

Advised by Dr. Tianyu Pang & Dr. Chao Du

May 2023 – Aug 2023

- Conducted evaluations on cross-modal models (e.g., Stable Diffusion, Whisper) to assess their consistency under a range of common data corruptions.
- Developed a rigorous benchmark for assessing the self-consistency of these models. The benchmark was designed to provide a comprehensive understanding of model behavior, incorporating a wide range of scenarios and inputs to measure their resilience and accuracy.

WORKSHOPS AND COMPETITIONS

- **CLAS 2024: The Competition for LLM and Agent Safety, NeurIPS 2024**
- **Secure and Safe Autonomous Driving (SSAD) Workshop and Challenge, CVPR 2023**

TEACHING

CS 307 - Modeling and Learning in Data Science (Spring 2022)

- Teaching Assistant with *Prof.* Bo Li and *Prof.* David Forsyth

SELECTED HONOR & AWARDS

Graduate Conference Travel Award	<i>March. 2023</i>
Accumulative Research Innovation & Entrepreneurship Index: 1st Department-wide	<i>June. 2021</i>
Meritorious Winner, MCM COMAP's Mathematical Contest in Modeling	<i>Apr. 2020</i>
First Prize, China Harbour Scholarship	<i>Jan. 2020</i>
First Prize, The Chinese Mathematics Competitions, Zhejiang Province	<i>Nov. 2018</i>

PROFESSIONAL SERVICE

Session Chair — ICLR 2025

Top Reviewer — NeurIPS 2025

Program Committee— ICML, NeurIPS, ICLR, CVPR, ECCV, AISTATS, AAAI, COLM, ACL, JIMR, etc.